

# Ravali N

## Senior Full Stack Engineer

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### PROFESSIONAL SUMMARY

Senior Full Stack Developer with 7+ years of extensive experience in designing and developing scalable, high-performance applications using Java, Spring Boot, and microservices architecture. Proven ability to build robust RESTful APIs, GraphQL services, and event-driven systems leveraging technologies like Kafka and RabbitMQ to enable seamless communication across distributed systems. Strong expertise in frontend development using React and modern JavaScript frameworks to create responsive and user-friendly interfaces. Adept at working with both relational and NoSQL databases, with a focus on query optimization and performance tuning to support high-volume applications. Hands-on experience with cloud-native development using AWS services such as Lambda, EKS, and API Gateway, along with containerization and orchestration using Docker and Kubernetes. Experienced in implementing and managing CI/CD pipelines using Jenkins, GitLab CI, and Harness to ensure efficient and reliable deployments. A collaborative and results-driven professional who works effectively across teams to deliver end-to-end solutions, integrating frontend, backend, and cloud components to meet business and technical requirements.

### TECHNICAL SKILLS

|                                            |                                                                                                                                                                                                                                   |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>              | Java, JavaScript, Python, C, C++, C#, TypeScript, Golang                                                                                                                                                                          |
| <b>Backend Technologies:</b>               | Spring Boot, Spring MVC, Spring Security, Spring Cloud, Node.js, FastAPI, Flask, RESTful APIs, GraphQL, Microservices Architecture, Event-Driven Systems                                                                          |
| <b>Frontend Technologies:</b>              | React.js (Hooks, Context API, Redux), Angular, HTML5, CSS3, JavaScript (ES6+), TypeScript, Bootstrap, Material UI, Responsive Design.                                                                                             |
| <b>Databases and Data Technologies:</b>    | MySQL, PostgreSQL, SQL Server, DB2, MongoDB, DynamoDB, Redis, Data Modeling, Query Optimization, Indexing                                                                                                                         |
| <b>Cloud Platforms and Data Platforms:</b> | AWS (Lambda, EKS, S3, API Gateway, CloudWatch, IAM), Azure (App Services, Azure Functions, AKS, Azure DevOps, Storage Accounts), Docker, Kubernetes, Terraform, Helm, CI/CD (Jenkins, GitLab CI, Harness), Infrastructure as Code |
| <b>Security:</b>                           | OAuth2, JWT, RBAC, Spring Security, Secure API Design, Authentication & Authorization                                                                                                                                             |
| <b>API and Integration:</b>                | OpenAPI (Swagger), API Gateway Integration, Third-Party API Integration, Web Services (REST & SOAP), JSON/XML, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, CI/CD Pipelines, Helm, Git, Version Control                |
| <b>Tools &amp; Platforms:</b>              | Git, Maven, Gradle, SonarQube, Postman, IntelliJ IDEA, VS Code, Jira, Confluence                                                                                                                                                  |
| <b>Testing and Monitoring:</b>             | JUnit, PyTest, Integration Testing, Unit Testing, Postman, Code Quality (SonarQube), Logging and Monitoring                                                                                                                       |

### WORK EXPERIENCE

#### Senior Full Stack Developer

Dec 2024 – Present

#### Walmart, Dallas, Tx

##### Responsibilities:

- Collaborating with internal technical teams to identify and develop platform elements required to support key wireless product features in a timely and efficient manner.
- Designed and developed RESTful APIs using Java 17 and Spring Boot 3.x following OpenAPI 3.0 specifications to ensure standardized API contracts and seamless integration across services.
- Built scalable microservices using Spring Cloud, implementing service discovery, centralized configuration, and resilient inter-service communication.
- Developed complex database integrations using Spring Data JPA and Hibernate with SQL Server, writing optimized T-SQL queries, stored procedures, and indexing strategies to improve query performance.
- Reduced API latency by resolving N+1 query issues, implementing caching strategies, and tuning Hibernate fetch plans.
- Implemented complex business logic using Java collections, generics, streams, and lambda expressions to efficiently process large volumes of data.

- Integrated Java applications with relational databases using JDBC, managing transactions, connection handling, and optimized query execution.
- Built multithreaded components using Java concurrency utilities to handle parallel processing, thread synchronization, and safe resource sharing in high-load environments.
- Developed high-performance, responsive web applications using JavaScript and modern UI frameworks.
- Designed and developed RESTful APIs and microservices using Java, Spring Boot, and Spring MVC, enabling seamless inter-service communication.
- Implemented Spring Security with JWT and OAuth2 for authentication and authorization, securing APIs against unauthorized access.
- Migrated legacy Java applications to Spring Boot microservices architecture, improving maintainability and scalability.
- Built and optimized transaction-based applications leveraging Spring Boot, Hibernate/JPA, and relational databases (MySQL, PostgreSQL, DB2).
- Developed a smart TV application from scratch using JavaScript and React, delivering an intuitive and responsive user interface optimized for large-screen experiences.
- Collaborated on backend components using Python, integrating AI-powered voice recognition capabilities to enable seamless voice-based movie search across multiple streaming apps.
- Implemented a scheduler feature for automated TV power ON/OFF functionality, improving user convenience and energy efficiency.
- Contributed to end-to-end system design, working across frontend and backend layers to ensure seamless interaction between user commands and device controls.
- Collaborated with cross-functional teams to integrate voice services and AI modules, ensuring smooth data flow and minimal latency across the TV ecosystem.
- Designed pixel-perfect UI layouts using CSS, Bootstrap, and Flexbox, ensuring consistency across devices and browsers.
- Developed Python-based automation scripts for cloud resource management, utilizing AWS SDK (Boto3) to dynamically provision and manage cloud infrastructure resources.
- Enhanced the performance of NoSQL databases like MongoDB by optimizing queries and indexing strategies, which reduced read/write latency for high-volume applications.
- Utilized Terraform to create Infrastructure-as-Code (IaC) for provisioning cloud resources in AWS and Google Cloud, automating repeatable tasks and increasing efficiency.
- Worked with Docker and Kubernetes to containerize a set of legacy applications, improving portability and scalability across development, staging, and production environments.
- Built RESTful APIs using Java and Flask/FastAPI to support seamless communication between services in a microservices architecture.
- Integrated third-party APIs and internal services using Python to enhance system capabilities and streamline workflows.
- Developed unit and integration tests using pytest and unittest frameworks to ensure high code quality and reliability in production deployments.
- Created a GraphQL API for an internal tool, enabling dynamic querying and data manipulation with minimal changes to the existing infrastructure.
- Developing and maintaining core platform services using AWS, Kubernetes, Docker, Helm, and Lambda by keeping yourself up to date with the latest technologies and industry trends.
- Implemented event-driven architecture using Confluent Kafka with Spring for Apache Kafka to support real-time event streaming and asynchronous processing.
- Designed and implemented RabbitMQ-based messaging solutions using Spring AMQP to handle queue-based workflows and background processing.
- Developed and enforced schema governance policies to ensure consistency and quality in GraphQL schemas.
- Built containerized Spring Boot applications with Docker, deployed to Kubernetes (EKS), ensuring scalability and high availability.
- Integrated CI/CD pipelines with Jenkins and GitLab CI for automated builds, code quality checks (SonarQube), and deployment.
- Utilized Spring MVC, Dependency Injection, and JSTL to handle presentation and business logic, and validate UI data using JSF Validators, jQuery, TypeScript and JavaScript
- Implemented solutions using latest AWS technologies, including AWS CDK, to automate infrastructure deployments and enhance system reliability.
- Experience in creating unit and integration tests using JUnit, SOAP UI, and Postman for testing and verifying the functionality of software applications.
- Collaborated on backend components using Python, integrating AI-powered voice recognition capabilities to enable seamless voice-based movie search across multiple streaming apps.

**Dish Network, Plano, TX****Responsibilities:**

- Built and maintained RESTful APIs and SOAP-based web services using Java, Spring Boot, and JAX-RS for seamless communication between microservices.
- Developed enterprise-grade applications using Java 17 and Spring framework, integrating with databases, messaging systems, and caching layers.
- Led the migration of a monolithic application to microservices, breaking it down into smaller, decoupled services that interact via RESTful APIs, and deploying it on cloud infrastructure with Terraform.
- Performed JVM tuning, thread optimization, and garbage collection analysis to enhance performance.
- Created Swagger/OpenAPI documentation for API standardization and integration consistency.
- Implemented microservices architecture with Spring Boot, ensuring modularity, scalability, and improved maintainability of backend services.
- Performed JVM tuning, thread optimization, and garbage collection analysis to enhance application performance.
- Implemented logging, monitoring, and health checks using Spring Boot Actuator and centralized monitoring tools.
- Designed and developed scalable, high-performance backend systems using Java, implementing object-oriented design principles and design patterns.
- Worked with relational databases such as PostgreSQL, MySQL, and Microsoft SQL Server, writing complex SQL queries, stored procedures, and optimizing database interactions.
- Enabling the implementation of cloud-based application deployment to ensure reliability, resiliency, and observability of the application.
- Implemented a hybrid deployment model, running services concurrently on Amazon EKS and AWS Lambda, to validate performance benchmarks before full serverless migration.
- Re-engineered legacy EKS-hosted microservices to be stateless and event-driven, enabling smooth transition to Lambda functions with minimal downtime.
- Created modular and reusable Lambda layers to share common code (e.g., auth, logging, utilities) across multiple serverless functions.
- Integrated Lambda functions into existing API Gateway endpoints and updated routing logic to support both Kubernetes-based and serverless handlers.
- Optimized build and deployment processes using Harness pipelines, enabling automated rollouts, canary releases, and rollback strategies across EKS and Lambda environments.
- Migrated critical services from EKS-based deployments to AWS Lambda, enhancing scalability and reducing infrastructure overhead by adopting a serverless architecture.
- Developed and maintained application code to be adaptable for both EKS and AWS Lambda deployments, ensuring seamless execution across different environments.
- Built and managed CI/CD pipelines using Harness to automate deployments to both EKS clusters and AWS Lambda, streamlining release processes.
- Refactored multiple codebases to support event-driven serverless execution, enabling efficient integration with AWS services such as API Gateway, S3, and DynamoDB.
- Collaborated with cross-functional teams to update and validate multiple applications for compatibility with AWS Lambda, ensuring functionality and performance parity with Kubernetes-based services.
- Optimized Java applications for performance, utilizing JVM tuning, garbage collection optimizations, and efficient multi-threading.
- Developed and integrated secure authentication and authorization mechanisms using OAuth 2.0, JWT, and Spring Security.
- Implemented secure authentication and authorization using Spring Security, OAuth 2.0, JWT, and role-based access control (RBAC) to protect APIs and applications.
- Worked with relational databases such as PostgreSQL, MySQL, and Microsoft SQL Server, writing complex SQL queries, stored procedures, and optimizing database interactions.
- Worked on Kubernetes cluster as a part of deployment activity. Familiar with OpenShift commands mainly for deployment.
- Utilized Spring/MVC framework and J2EE/JSON technology to enable interactions between the view layer and implemented different design patterns.
- Implemented spring boot microservices to process messages into Kafka cluster and used Hibernate for persistence and transaction management.
- Integrated Spring and Hibernate for backend components and services, and established database connectivity using JDBC And Spring ORM for Oracle.
- Contributing to the evolution of the current engineering strategy and development process by proposing innovative ideas and new technologies that can improve the development process and enhance the overall

quality of the software product.

- Enhanced legacy application by building new components in Angular JS and Typescript.
- Implemented features like Actuator in Spring boot for monitoring and tracing my application. Developed controller classes, complex services, database integration and backend process.
- Also created YAMEL files for using a swagger API within the UI.
- Collaborating with DevSecOps to follow CI/CD pipeline practices and ensure the quality and security of the code by incorporating security practicefs in the software development process.
- Enabling the implementation of cloud-based application deployment to ensure reliability, resiliency, and observability of the application.
- Built backend services using Python (FastAPI/Flask) for data processing, API development, and integration with enterprise systems.
- Contributed to AI-driven automation workflows to streamline development and reduce manual intervention.
- Collaborated with cross-functional teams to deliver end-to-end solutions aligned with business requirements.

## **Full Stack Developer**

**Jan 2022 – Dec 2022**

**MetLife, New York, NY**

### **Responsibilities:**

- Gathered business and functional requirements, and designed proof of concept with Spring Boot.
- Strong expertise in developing modern, responsive, and accessible web applications using JavaScript (ES6+), TypeScript, and CSS
- Utilized Java for backend logic and data validation in a cloud-native environment (AWS Lambda, S3, EC2).
- Created custom Python utilities to parse logs, monitor services, and generate automated reports for stakeholders.
- Collaborated with DevOps teams to deploy Java applications using CI/CD pipelines (GitLab CI, Jenkins) and Docker containers.
- Optimized Java code for performance and scalability, reducing script execution times and resource usage.
- Wrote reusable Java modules and maintained documentation to support ongoing team development and onboarding.
- Proficient in using Bootstrap components, themes, and utilities for rapid UI prototyping and consistent styling
- Extensive experience with frontend frameworks like React (Hooks, Context API, Redux), Angular (RxJS, NgRx, Services), and Vue.js (Vuex, Composition API)
- Experience in creating custom, reusable UI components and integrating design systems such as Material UI and Ant Design
- Hands-on experience in developing single-page applications (SPAs) with dynamic routing and client-side state management
- Proficient in integrating RESTful APIs and WebSockets for real-time data updates in frontend applications
- Strong understanding of DOM manipulation, event handling, and optimizing reflows/repaints for better UI performance
- Expertise in lazy loading, code splitting, and tree shaking to improve frontend performance and reduce page load times
- Hands-on experience in writing unit and integration tests using Jest, Mocha, Chai, Cypress, and Selenium
- Implemented progressive web applications (PWAs) with offline capabilities using service workers and caching strategies
- Strong experience in customizing Bootstrap themes, implementing dark/light mode, and applying animations with CSS and JavaScript
- Developed components using J2EE principles and design patterns, including Singleton, Factory, and Data Access Object.
- Utilized Spring MVC, Dependency Injection, and JSTL to handle presentation and business logic, and validate UI data using JSF Validators, jQuery, TypeScript and JavaScript.
- Implemented Web Services using WSDL and SOAP protocol, and utilized Spring-AOP for logging and user session validation.
- Developed Action class and Action Form for business logic using spring framework, and integrated Spring DAO for data access with Hibernate.
- Optimized GraphQL query performance by implementing query batching and caching strategies using Apollo Client/Server
- Created Hibernate configuration and mapping XML files for database access.
- Influenced and guided technically proficient customers and partners to [specific outcomes].
- Utilized Azure App Services to host and manage web applications, enabling automatic scaling and load balancing to ensure optimal performance during peak usage.
- Developed and deployed Azure Functions for serverless computing, reducing operational overhead and improving the efficiency of event-driven tasks.
- Integrated Azure DevOps for continuous integration and continuous delivery (CI/CD), automating build and deployment pipelines to streamline development workflows.
- Embraced a bold and ambitious mindset to drive innovation and think big in all projects.
- Experience in creating unit and integration tests using JUnit, SOAP UI, and Postman for testing and verifying the functionality

of software applications.

- Troubleshoot and resolved production issues by analyzing logs, debugging code, and implementing fixes to maintain system stability and uptime.
- Created custom Python utilities to parse logs, monitor services, and generate automated reports for stakeholders.

## Software Developer

March 2019 – July 2021

### Analogue IT Solutions, Hyderabad, IN

#### Responsibilities:

- Developed enterprise-level full-stack applications using Java, Python, and JavaScript technologies, supporting business-critical workflows and data processing systems.
- Developed J2EE components like Java Beans, XML, Collection Framework, Servlets, JSP, JMS, and JDBC and deployed them in IBM WebSphere Server.
- Used Eclipse IDE for creating various J2EE artifacts like Servlets, JSPs, and XML.
- Implemented state management using Cookies, Session, and Application objects for client-side and server-side.
- Developed views and controllers for client and manager modules using Spring MVC and Core.
- Implemented business logic using Spring Core and Hibernate and performed data operations using Spring ORM with Hibernate.
- Automated deployment and configuration workflows using Python scripts, reducing setup time across environments.
- Created Python-based cron jobs and schedulers for routine data sync, monitoring, and alerting tasks.
- Parsed and analyzed large-scale JSON, XML, and CSV files using Python to support business intelligence workflows.
- Leveraged Python's multiprocessing and asyncio libraries to build high-performance, concurrent applications.
- Designed secure authentication and authorization logic in Java APIs using JWT and OAuth2 standards.
- Developed the action classes, Action form classes, created JSP's using Struts tag libraries and configured in Struts-config.xml, web.xml files.
- Developed Exception handling framework using log4j for logging and performed Unit test with JUnit.
- Implemented Exception handling code to recover from DB-specific, application-specific, business, and non-business logic specific exceptions.
- Performed back-end testing on Oracle and SQL Server databases by writing SQL queries to ensure correct content display on the UI as per requirements.
- Collaborated effectively with engineers, product managers, and leaders to innovate and deliver cutting-edge products.
- Designed and developed highly distributed system components, enhancing system performance/scalability/efficiency.
- Experience in creating unit and integration tests using JUnit, SOAP UI, and Postman for testing and verifying the functionality of software applications.
- Improved system performance by optimizing backend logic, reducing processing time, and enhancing database query efficiency.
- Developed and executed unit tests using JUnit and other testing frameworks to ensure code quality and reliability.
- Participated in code reviews and provided feedback to improve coding standards and maintainability of the codebase.
- Assisted in deployment and release management processes, ensuring smooth transitions between development, testing, and production environments.
- Diagnosed and resolved production issues by analyzing logs and debugging application components, ensuring minimal downtime.
- Collaborated with program managers, policy analysts, and IT teams to translate business requirements into scalable

## PROJECT

- Developed and maintained applications which are currently being used by the school.
- Developed an Attendance Tracking System (ATS) web application using Spring Boot, Spring Cloud, ReactJS, and MongoDB.
- Reduced manual attendance management efforts by automating processes with the ATS web application
- Ensured code quality and maintainability by implementing robust REST APIs and comprehensive unit tests.
- Achieved a 90% test coverage by using Mockito and JUnit for unit and integration tests.
- Optimized database queries and data models, noticeably reducing query execution time.
- Improved database performance, enhancing overall application responsiveness and efficiency.
- Reverse-engineered and refactored a Progressive Web Application (PWA) to boost its performance and maintainability.
- Applied real-time big data processing techniques using Kafka for efficient data streaming and processing.
- Improved data accessibility and latency by leveraging AWS cloud services for scalable data storage and processing.
- Implemented microservices architecture with Spring Cloud to ensure scalable and modular application design.
- Enhanced application scalability and reliability by leveraging cloud-based services and infrastructure.
- Conducted thorough code reviews and implemented best practices for clean and maintainable code.

- Collaborated with cross-functional teams to align software development efforts with project goals.
- Implemented Maven for project build and dependency management, ensuring smooth and consistent builds.

## **EDUCATION**

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### **Bachelor of Engineering**

Samskruthi Institute of Science and Technology. Hyderabad, India

### **Master of Science**

Northwest Missouri State University. Missouri, USA